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Kumaraswamy Exponentiated Odds Ratio Family of Distributions

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MOTIVATION

The Exponentiated Odds Ratio generator proposed by Chen et al. [2] offers a range of potential distribution families. Focusing on the Weibull-G family of distributions as proposed by Bourguignon et al. [1] the CDF of the baseline distribution is given, $G(x, \psi)$. Our research explored the Kumaraswamy-G family of distribution as our baseline function. The CDF is computed as:

$$F(x) = 1 - [1 - G^a(x)]^b$$

- Traditional models lack flexibility with the analysis of modern datasets.
- The new family offers a wide variety of distribution possibilities exceling in performance compared to traditional distributions.

SUBFAMILY

The Kumaraswamy-G family offers multiple subfamilies of distributions, this study evaluated the Type-2 Gumbel Inverse Kumaraswamy (TIK) distribution. The CDF and PDF are represented below by F_{T2GS} and f_{T2GS} respectively.

$$F_{T2GS}(x) = e^{-\left(\alpha(1-G^a(x))^{-b-1}\right)^{\beta}}$$

$$f_{T2GS}(x) = \frac{ab\beta}{\alpha} G^{a-1}(x)g(x) \dots$$

$$[1 - G^a(x)]^{-b-1}(-1)^{-\beta-1} \dots$$

$$e^{-\left(\alpha(1-G^a(x))^{-b-1}\right)^{\beta}}$$

- Further analysis was completed to obtain the distributions mathematical and statistical properties, including hazard rate functions, quantile functions, moments, entropy, and order statistics.
- Simulation studies and analysis of real-world datasets were completed to determine the properties and effectiveness of the proposed distribution.

HAZARD RATE FUNCTION

This function gives the specific probability an event will occur for any given time x . The Hazard rate function is also described as the failure rate function and is typically used in risk analysis. The hazard rate function for the proposed TIK distribution is given below. This function was used in the analysis of our distribution by various simulation studies.

$$h(x) = \frac{\beta}{\alpha} \cdot \frac{((1 - G(x, \psi)^a)^b)^{\beta+1}}{1 - (1 - G(x, \psi)^a)^b} \cdot \frac{ba g(x, \psi) G(x, \psi)^{a-1}}{(1 - G(x, \psi)^a)^{b+1}} \cdot \left[e^{\left[\frac{1 - (1 - G(x, \psi)^a)^b}{\alpha(1 - (1 - G(x, \psi)^a)^b)^{\beta}} \right]} - 1 \right]^{-1}$$

SPECIAL CASES

Uniform, exponential, and Weibull cases of the proposed probability function were explored. The uniform distribution is displayed in Figure 2. The pdf (left) exhibits a variety of behaviors including unimodal and bathtub configurations. The hazard rate plot (right) for the uniform distribution accentuates the bathtub shape as seen in the pdf plot.

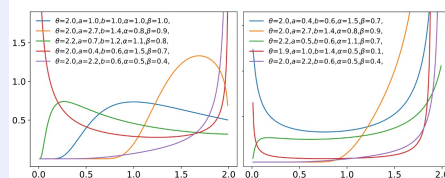


Figure 1. TIK-Uniform density and hazard rate plots given multiple parameters.

SIMULATION STUDIES

Simulation studies were also completed to further confirm methods of estimation for the TIK function. The error calculation is computed using the formula provided and the convergence plots for each parameter were graphed. Figure 2 shows the convergence graph for parameter a . Each parameter gave excellent parameter convergence allowing the distribution to be applied to real datasets.

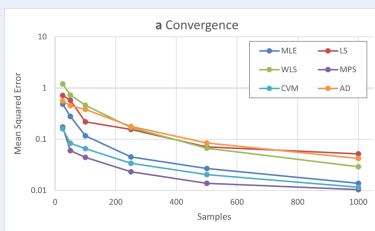


Figure 2. Convergence results for parameter a .

Error Calculation:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

APPLICATIONS

The applicability of the TIK distribution was proven with the assessment of three different datasets. The Araset dataset ($n=5$) plots are presented in figure 3. Further comparisons were also run, including various goodness of fit tests.

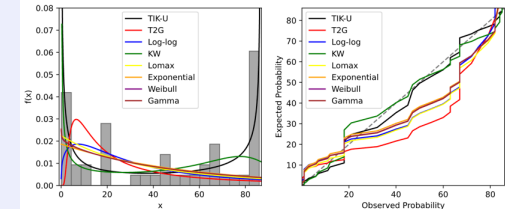


Figure 3. Histogram (left) with fitted PDF's and expected vs. observed (right).

CONCLUSIONS

The proposed TIK distribution offers a solution to potential issues traditional distributions pose when modeling real world data. The rigorous statistical analysis completed determined the adaptability and versatility this distribution offers. This emphasizes the superiority of the distribution as well as the advancement in the field of statistical analysis.

REFERENCES

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